

Brainstorming: Multimedia Authoring Talk

Potential Talk Titles

- Escaping the Timeline: Semantic Structures for Multimedia Authoring
- Editing Meaning, Not Pixels: Designing Substrates for Human-AI Media Authoring
- The Representation Gap: Bridging Human Intent and AI Generation in Video

Part 1: Concrete Motivational Examples (The "Hook")

(See previous versions for the Webb Smith/Disney Storyboard story)

Part 2: Deconstructing the Storyboard (The 3 Principles)

(See previous versions for mapping Semantic Units, Shared Reference, and Iteration)

Part 3: The Transition (Speaker Notes & Slide Org)

(See previous versions for the "Representation Gap" timeline vs. storyboard transition)

Part 4: Narrative Arc - "Escaping the Timeline"

Chapter 1: Supporting Semantics

(See previous versions for the Lyric Video Deep Dive)

Chapter 2: Pinning the Fleeting Moment (Shared Reference)

(See previous versions for Snapstream -> VisPoll/Beyond Show of Hands transition)

Chapter 3: The Substrate for Iteration (Generative AI)

Slide: Introduction to Pillar 3

- **The Principle:** Iteration: Elevate compositional structures into active substrates to enable fluid reflection-in-action.
- **Speaker Notes (The Setup):** *"We've talked about semantics, and we've talked about shared references. Now, I want to talk about how we bring these together to support the most crucial part of authoring: Iteration. And I want to frame this in the context of our current era of Generative, Agentic AI."*

The Problem: The Myth of the "Black Box"

- **The Agentic AI Promise:**

chat interface. The pitch is: 'Just tell the AI what you want, and it will do the work for you.' It treats media authoring as an extraction problem—getting the idea out of your head and onto the screen as fast as possible."

- **The Walter Murch Counter-Argument:**

- *Speaker Notes:* "But the legendary film editor Walter Murch once posed a brilliant thought experiment. He asked: '*What if there were a machine—a black box—that could connect directly to a director's brain and translate their intentions instantly into a finished film?*' His conclusion? Most great directors wouldn't use it. Why? Because the *process* of creation is vital. The friction, the discovery, the reflection... iteration is where the art actually happens."

- **The HCI Theory:** Donald Schön's *Reflection-in-Action* (1983).

- *Speaker Notes:* "Donald Schön argued that creators don't just execute plans; they engage in a 'conversation with the materials.' Agentic AI chat boxes skip the conversation. So, how do we build AI that actually supports reflection and iteration?"

The Formative Study: How Experts Actually Iterate

- **The Empirical Evidence:**

- *Speaker Notes:* "To answer this, we interviewed 9 expert video creators across different genres. We found that while their specific workflows differed wildly, they all shared core, underlying stages."

- **The Core Stages & Intermediate Artifacts:**

- *Visual:* A diagram showing stages mapping to different tools:
 - 1. Asset Collection -> Moodboards / Canvas
 - 2. Narrative Development -> Text Documents / Scripts
 - 3. Scene Planning -> Spreadsheets / Audio-Video sync tables
 - 4. Pacing Adjustment -> NLE Timelines
- *Speaker Notes:* "At every single one of these stages, they didn't just stare at the raw video. They relied on **intermediate artifacts** and representations. They needed these artifacts to effectively inspect and reflect on different aspects of their content."

The Solution: Compositional Structures (Video Origami - CHI 2025)

- **Defining Compositional Structures:**

- *Speaker Notes:* "When we analyzed these artifacts, we realized they weren't just random UIs. They are what we call **Compositional Structures: structures that visualize the relationships between individual elements through certain aspects of the content.**"
- *Examples:*
 - A **Set** (for spatial/asset collection).
 - A **Linear List** (for narrative flow).
 - A **Table** (to synchronize visual and narrative).
 - A **Timeline** (to organize temporally).

- *Speaker Notes*: "In our recent work, *Video Origami*, we propose design guidelines for building human-AI co-creation systems around these structures. We argue that AI shouldn't bypass the structures; AI should be infused *into* them."
- **Going Deep: The Instantiation in Video Editing:**
 - *Speaker Notes*: "We instantiated these guidelines in a novel video editing environment. Instead of a single timeline, we give the user active, AI-synchronized compositional structures."
 - *Example (Fluid Iteration)*: "Because the AI understands the structures, iteration is fluid. If a user reflects on the pacing in the Timeline and physically shortens a clip, the AI propagates that intent, automatically suggesting a revised script in the Linear Narrative view. The intermediate artifacts are no longer dead documents; they are a synchronized, active substrate for co-creation."

Broadening the Scope

- *Speaker Notes*: "And this principle extends far beyond video. In fact, not only for video editing, but for writing code, designing slide decks, or architecting software—any complex creative work requires iteration. If we want AI to truly assist us, we must design systems that expose the compositional structures of those domains."

Conclusion

- *Brief Recap*: Bring back the Webb Smith bulletin board image. Remind the audience how it provided semantics, shared reference, and iteration.
- *Final Takeaway*: "The history of media authoring is a constant battle against the timeline and the pixel. As we build the next generation of creative tools, our goal should not just be to build better AI 'black boxes' that do the work for us. Our goal must be to build richer, more expressive interfaces that speak the structural language of the creator. Because when we allow users to author by structure and meaning, we don't just close the representation gap; we allow them to finally escape the timeline "